



Business Intelligence with Alibaba Cloud

by Mohamad Ikhsan Nurulloh

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- **Testing Only:** The information provided is for testing and development purposes only, not for production environments.
- **Refer to Official Documentation:** Always consult the official Alibaba Cloud documentation before implementing solutions in a production setting.
- **Proceed at Your Own Risk:** Use this guide at your own risk, being aware of potential deployment implications.
- **Security Considerations:** Configurations use default settings and open access for testing. Always follow Alibaba Cloud's security recommendations to protect your data and infrastructure.

What is Alibaba Cloud



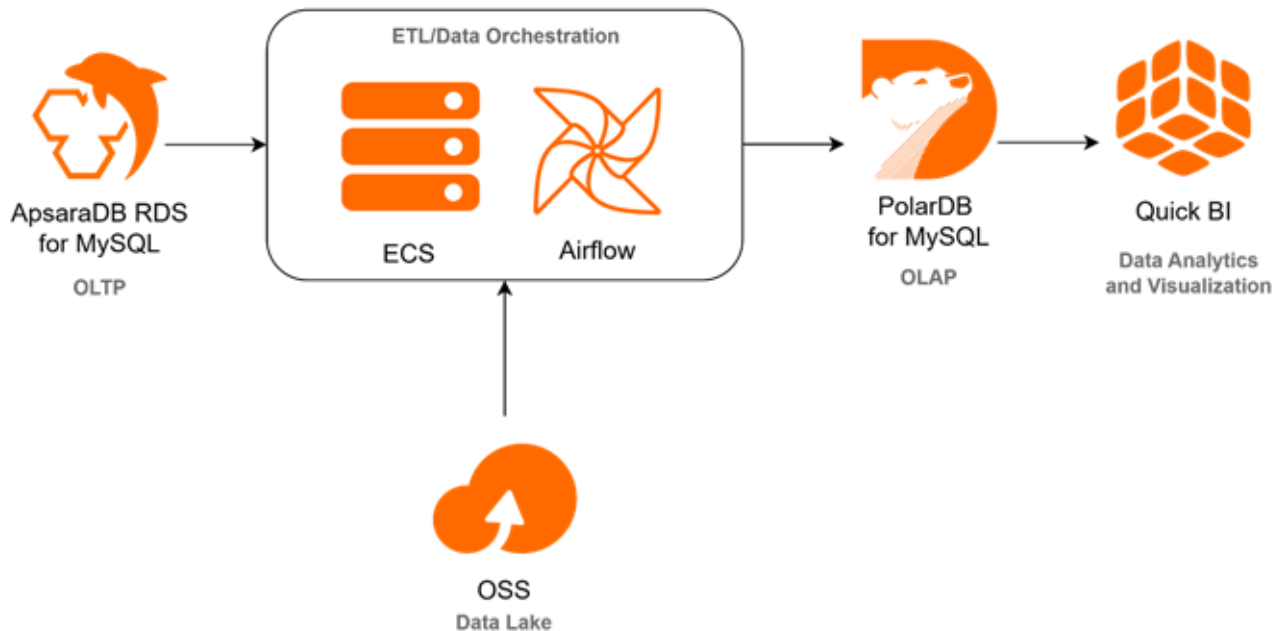
Alibaba Cloud is a leading cloud computing service provider that offers a wide range of cloud services, including computing, storage, and data analytics solutions. Its BI tools enable organizations to visualize data effectively, facilitating better insights and informed decisions.

Why Alibaba Cloud

- **User-Friendly Interface:** Alibaba Cloud's BI tools come with intuitive interfaces that allow users to create and share dashboards easily without requiring extensive technical knowledge.
- **Comprehensive Analytics Capabilities:** The platform supports complex queries and diverse data visualizations, making it suitable for various analytical needs.
- **Seamless Integration:** It integrates smoothly with various databases and data sources, enhancing the overall data management process.
- **Collaboration Features:** Users can easily share dashboards and reports, promoting teamwork and collaborative decision-making.

Alibaba Business Intelligence Architecture

Business Intelligence Architecture with Alibaba Cloud



Overall Architecture: The Business Intelligence architecture with Alibaba Cloud starts with data stored in ApsaraDB RDS for MySQL (OLTP) or OSS Data Lake. The data undergoes processing using ETL tools, particularly Airflow hosted on ECS. After processing, the data is analyzed in PolarDB for MySQL (OLAP) and then visualized using Quick BI.

Details Business Intelligence Architecture with Alibaba Cloud

- **ApsaraDB RDS for MySQL:** Optimized for Online Transaction Processing (OLTP).
- **Object Storage Service (OSS):** Functions as a data lake for storing large volumes of unstructured data.
- **Elastic Compute Service (ECS) hosted Airflow:** Used for orchestration and Extract, Transform, Load (ETL) processes.
- **PolarDB for MySQL:** Designed for Online Analytical Processing (OLAP), supporting complex queries and analytics.
- **Quick BI:** Provides data analytics and visualization capabilities to facilitate insights and reporting.

Setting Up Business Intelligence Infrastructure

Configuring ApsaraDB RDS for MySQL

The screenshot displays the Alibaba Cloud ApsaraDB RDS console configuration page for a MySQL instance. The interface is organized into several sections for configuration:

- Billing Method:** Options include Subscription, Pay-as-you-go, and Serverless. A note states that the backup feature is automatically enabled after creating an instance, with a free quota on backup storage.
- Region:** Set to Singapore.
- Database Engine:** MySQL 8.0 is selected. Other options include PostgreSQL and MariaDB.
- Microsoft SQL Server:** An option to select an engine is available.
- Edition:** Basic Edition and High-availability Edition are shown. The Cluster Edition is also visible.
- Product Type:** Standard and Yitian are shown. A note indicates that the Yitian product type is more cost-effective but is supported for a fewer instance types and regions than the standard product type.
- Storage Type:** ESSD, PL1, and General ESSD are shown. A checkbox for Cloud Disk Encryption is also present.
- Network Type:** VPC is selected.

On the right side, the pricing and subscription details are displayed:

- Quantity:** 1
- Subscription Duration:** 1 Month, 1 Year, and 5 Years are shown. The baseline price is \$109.16 USD/Month.
- Preferential Monthly Fee:** A table shows the monthly fee for different durations: 1 Month (\$109.16 USD/Month), 1 Year (\$76.42 USD/Month, 30% off the baseline price), and 5 Years (\$32.75 USD/Month, 70% off the baseline price).
- Auto-renewal:** A checkbox to enable auto-renewal is shown.
- Billing Method:** Subscription
- Region:** Singapore
- Database Engine:** mysql 8.0
- Edition:** Cluster Edition
- Product Type:** Standard
- Price:** \$0.00 USD (View Details)
- Pay Now:** A button to proceed with the purchase.

The configuration details for ApsaraDB RDS for MySQL on Alibaba Cloud encompass various aspects, including billing methods, database engines, and network types

Provisioning ApsaraDB RDS for MySQL Instances

The screenshot displays the Alibaba Cloud console interface for an ApsaraDB RDS instance. The top navigation bar includes the Alibaba Cloud logo, a search bar, and links to various services like Expenses, ICP, Enterprise, Support, Tickets, and a user profile. The breadcrumb trail indicates the path: ApsaraDB RDS / Instances / Basic Information. The instance identifier is 'rm-gs5i9lhj03bys56y2', and its status is 'Running'. A sidebar on the left lists various management options such as Basic Information, Accounts, Databases, Whitelist and SecGroup, Database Connection, Database Proxy, Monitoring and Alerts, Backup and Restoration, Data Security, Service Availability, Logs, Parameters, Data Migration and Synchronization, and Autonomy Services. The main content area is divided into three sections: Basic Information, Instance Distribution, and Status. The Basic Information section provides details about the instance ID, resource group ID, tags, type and edition, storage type, and database proxy. The Instance Distribution section shows the disaster recovery status and the number of instances. The Status section displays the current status (Running), billing method (Subscription), and expiration date.

Basic Information

Instance ID	rm-gs5i9lhj03bys56y2	Name	rm-gs5i9lhj03bys56y2
Resource Group ID	rg-acfmuwktf632koi	Resource Group Name	Default Resource Group
Tags	acscrmcgld : rg-acfmuwktf632koi	Region and Zone	Singapore ZoneB(Primary)
Type and Edition	Regular Instance(Cluster Edition)	Network Type	VPC View Details
Storage Type	PL1 ESSD Upgrade Storage Type	Product Type	Standard
Database Proxy	View Details		

Instance Distribution

Disaster Recovery	0 Add	Analytic Instance	0 ApsaraDB for ClickHouse Instance
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Status

Status	Running	Billing Method	Subscription (Auto-renew Disabled) Change to Pay-As-You-Go
Created At	Nov 1, 2024, 20:52:14	Expired At	Dec 1, 2024, 23:00:00

This section summarizes the successful provisioning of an ApsaraDB RDS for MySQL instance, detailing critical information such as instance ID, resource group ID, and management options available for efficient database operation and monitoring.

Mastering DMS SQL Console

The screenshot displays the DMS SQL Console interface. The top navigation bar includes 'Home', 'Scene Guide', and tabs for 'Product: oltp' and 'Product: olap'. The main area is divided into a left sidebar with 'Frequently Used Features' and 'Database Instances', a central SQL editor, and a bottom results panel.

The SQL editor shows a query: `SELECT * FROM 'superstore_oltp'`. The results panel displays a table with 7 rows and 7 columns: No., category, city, country, customer_id, customer_name, and manufacturer.

No.	category	city	country	customer_id	customer_name	manufacturer
1	Furniture	Henderson	United States	CG-12520	Claire Gule	Bush
2	Furniture	Henderson	United States	CG-12520	Claire Gule	Hon
3	Office Supplies	Los Angeles	United States	DV-13045	Darrin Van Huff	Universal
4	Furniture	Fort Lauderdale	United States	SO-20335	Sean O'Donnell	Bretford
5	Office Supplies	Fort Lauderdale	United States	SO-20335	Sean O'Donnell	Eldon
6	Furniture	Los Angeles	United States	BH-11710	Brosina Hoffman	Eldon
7	Office Supplies	Los Angeles	United States	BH-11710	Brosina Hoffman	Newell

The bottom status bar indicates: 'The execution is successful. 3000 rows are returned. The total execution duration is 212ms. Up to 3000 rows can be returned by a single query. To obtain more data in a single query, use Data export.'

An overview of the DMS SQL Console for ApsaraDB RDS is provided, highlighting navigation features and sample SQL query execution for easy database management.

Configuring PolarDB RDS for MySQL

Alibaba Cloud

Cart Tickets Bills English [ikhsannur****@...](#)

PolarDB

[Product Details](#) [Documentation](#) [Console](#)

If you have consistent resource requirements and do not need to scale out clusters within a year, we recommend that you use subscription clusters. If your resource requirements occasionally change and you need to scale out clusters monthly or even weekly, we recommend that you use pay-as-you-go clusters together with compute plans. Compute plans can be shared among clusters. You can also purchase storage plans to reduce storage costs.

Product Type

☒ Subscription ☐ Pay-as-you-go ☐ Compute Plans ☐ Storage Plans ☐ Serverless

Creation Method

Create a brand new PolarDB

Region

☐ China ☐ Asia Pacific ☒ Europe and America

The geographical location where the cluster is deployed. It cannot be modified after purchase. Make sure that the PolarDB cluster is in the same region as the ECS instance. Otherwise, the PolarDB cluster and the ECS instance cannot be connected over the internal network.

Primary Zone

You can deploy your PolarDB cluster and the ECS instance in the same zone or in different zones.

Network Type

Please make sure that the ECS you want to connect to is on the same VPC network. To use another VPC network or VPC switch, please create it in the [console](#).

VPC

Default VPC

VSwitch

Default VSwitch

Amount Payable **\$0.00**
\$164.73 [View Detail](#)

Configuration options for PolarDB RDS on Alibaba Cloud are discussed, including product types and geographical deployment regions for flexible cloud database management.

PolarDB RDS for MySQL Instance Overview

The screenshot displays the Alibaba Cloud PolarDB console interface. At the top, the navigation bar includes the Alibaba Cloud logo, a search bar, and various service links like Expenses, ICP, Enterprise, Support, and Tickets. The main header shows the instance ID 'pc-f2z2u9o65c7751952' in a 'Running' state, with buttons for 'Log on to Database', 'Data Transmission', 'Restore Databases/Tables', and 'Refresh'.

The left sidebar contains a navigation menu with categories like Basic Information, Performance Monitoring, Settings and Management, Security, Accounts, Databases, Service Availability, Data Lifecycle, Backup and Restoration, Parameters, Version Management, Binary Logs, Diagnostics and Optimization, Quick Diagnostics, Log Management, Slow SQL, Logs and Audit, SQL Explorer and Audit, and Security Audit.

The main content area is titled 'Basic Information' and displays a table of instance details:

Property	Value
Region	UK (London)
Compute Billing Method	Subscription Switch to Pay-as-you-go Unsubscribe (Release)
Default VSwitch	vsw-d7ows8kuxwhdunnt7jd Switch
Compatibility	100% compatible with MySQL 8.0 Switch
GDN	Create GDN
Auto-renew	Off On
Kernel Version	8.0.1.1.46
Storage Type	PSLS
Resource Group ID	rg-achwskts532koi
Tags	Add Tag
Default VPC	vpc-d7oactidvlekhwhdmonjg Switch
Zones	London Zone A
Created At	Nov 1, 2024, 21:33:12
Maintenance Window	01:00-02:00 (UTC+07:00) Modify
Edition	Cluster Edition
Expiration Time	Dec 1, 2024, 23:00:00
Database Type	General Specification
Storage Payment Method	Subscription Change Storage Billing Method to Pay-as-you-go
Resource Group Name	default resource group

Below the table, there are two sections: 'Whitelists and Accounts' and 'Database Connections'. The 'Whitelists and Accounts' section includes a 'Connection Diagnosis' button and a message: 'To connect to and use the PolarDB cluster, you must configure an IP whitelist and create an initial account for the cluster. For more information, see [Configure IP Whitelist and Create Database Account](#).' It also shows 'Access Whitelist' (1 configured) and 'Cluster Account' (1 created). The 'Database Connections' section shows 'PolarProxy-2 cores + (0.5 to 32) PCUs' and includes buttons for 'Restart Database Proxy' and 'Database Proxy Settings'.

Detailed information about a specific PolarDB cluster instance is presented, including region, billing method, and management options for effective oversight.

Navigating PolarDB DMS SQL for MySQL Console

The screenshot displays the PolarDB DMS SQL for MySQL Console interface. The top navigation bar includes a Home button, a Scene Guide, and two product tabs: 'olap' and 'pc-f2z2u9o65c77'. The left sidebar shows a 'Frequently Used Features' section and a 'Database Instances' section. The main area is divided into a 'Database' pane on the left, a 'Workspace' pane in the center, and an 'Execution History' pane at the bottom.

The 'Database' pane shows a tree view of the database structure, including 'information_schema', 'mysql', 'olap', 'Table(3)', 'dm_superstore', 'peoples', 'superstore', 'View(3)', 'Procedure(3)', 'Function(3)', 'Trigger(3)', 'Event(3)', 'performance_schema', 'sys', and '___recycle_bin___'. The 'Workspace' pane shows a SQL query: `SELECT * FROM 'superstore'`. The 'Execution History' pane shows a table of execution results.

No.	category	city	country	customer_id	customer_name	manufacturer
1	Furniture	Henderson	United States	CG-12520	Claire Gute	Bush
2	Furniture	Henderson	United States	CG-12520	Claire Gute	Hon
3	Office Supplies	Los Angeles	United States	DV-13045	Darin Van Huff	Universal
4	Furniture	Fort Lauderdale	United States	SO-20335	Sean O'Donnell	Bretford
5	Office Supplies	Fort Lauderdale	United States	SO-20335	Sean O'Donnell	Eldon
6	Furniture	Los Angeles	United States	BH-11710	Brosina Hoffman	Eldon
7	Office Supplies	Los Angeles	United States	BH-11710	Brosina Hoffman	Newell
8	Technology	Los Angeles	United States	BH-11710	Brosina Hoffman	Mitel

The bottom status bar indicates that the execution is successful, 3000 rows are returned, and the total execution duration is 1334ms. It also provides a link to 'Data export'.

The SQL console interface for PolarDB RDS is showcased, illustrating navigation through databases and executing queries with user-friendly features.

Creating Object Storage Service (OSS) Bucket

Object Storage Service (OSS)

Overview

Buckets

Access Points

Object FC Access Points

Favorite Paths

Historical Paths

Resource Usage

Resource Plans

Health Detection

Data Service

Data Replication

Data Import

Anti-DDoS Pro

OSS Accelerator

Block Public Access

Self-service Tools

Basic Data

The data on this page is not updated in real time. A late billed based on different methods. [Learn More](#).

Storage Usage: Total (Excluding ECS Snapshots)

Actual Storage Usage: 0 Byte

Billed Storage Usage: 0 Byte

Compare (with Previous Month)

Common Tools

Data Migration to OSS

Supports online migration and offline migration.

[Online Migration](#) [Data Transport](#) [OSS](#)

Common Features

Versioning

Stores previous versions of objects and prevents unintended overwrites and deletions of your data. You can configure the validity period of previous versions by using lifecycle rules.

Create Bucket

Basic Information

* Bucket Name: supertorebucket (16/63) ✓

After you create a bucket, you cannot change the name of the bucket.

* Region: Indonesia (Jakarta)

After you create a bucket, you cannot change the region of the bucket. Alibaba Cloud services in the same region can access each other over the internal network.

Endpoint: oss-api-southeast-5.aliyuncs.com

* Storage Class: Standard | IA | Archive | Cold Archive

High-performance, highly reliable, and highly available storage class. This storage class is suitable for data that is frequently accessed.

* Redundancy Type: LRS | ZRS (Recommended)

After you enable ZRS for a bucket, you cannot convert the redundancy type of the bucket to LRS. The generated storage fees can only be offset by ZRS storage plans. [Learn More](#)

ZRS uses the multi-zone mechanism to distribute user data across three zones within the same region. If one zone becomes unavailable, the data that is stored in the other two zones can still be accessed. This ensures data availability. Take note that you are charged higher fees for ZRS compared with LRS. [Learn More](#)

Block Public Access: Not Activated

Block Public Access allows enterprises to easily limit public access to OSS resources in a centralized manner. This reduces the risk of unauthorized access and data leakage. [Learn More](#)

* ACL: Private | Public Read | Public Read/Write

All users, including anonymous users, can perform read and write operations on the objects in the bucket.

Description: Public Read/Write allows unauthorized users to read data in your bucket. This results in high security risks.

Billing Note: You may be charged for unexpected traffic over the Internet.

[OK](#) [Cancel](#)

The process of creating a new bucket in Alibaba Cloud's OSS is explained, detailing configuration options such as bucket name, region selection, and access control settings.

Advanced OSS Bucket Management

The screenshot displays the Alibaba Cloud OSS console interface for managing the 'superstorebucket' in the 'Indonesia (Jakarta)' region. The left sidebar provides navigation options for the Object Storage Service (OSS), including Overview, Buckets, Access Points, and Favorite Paths. The main content area shows the 'Objects' section, which includes a search bar, a table of objects, and a list of actions. The table lists objects with columns for File Name, Object Size, Storage Class, Updated At, and Actions. The object 'peoples.csv' is highlighted, showing a size of 0.108KB and a storage class of Standard. The Actions column includes links for 'View Details' and a list of actions: 'Set Object Metadata', 'Export URL List', 'Download', 'Restore', and 'Permanently Delete'. The page also features a 'Task List' button and a 'Feedback' link.

File Name	Object Size	Storage Class	Updated At	Actions
peoples.csv	0.108KB	Standard	Nov 2, 2024, 14:04:40	View Details

Management of an OSS bucket named "superstorebucket" is focused on, detailing navigation options and functionalities for efficient object management.

Configuring ECS Instances

The screenshot displays the Alibaba Cloud ECS Custom Launch configuration interface. The top navigation bar includes the Alibaba Cloud logo, a search bar, and links for Workbench, All Resources, and UK (London). The main header shows 'Elastic Compute Service' with tabs for 'Quick Launch' and 'Custom Launch'. The right sidebar contains links for Savings Plans, Pricing, Buy Disk, and Console.

Billing Method: Subscription (selected), Pay-as-you-go, Preemptible Instance.

Region: China (Hong Kong). A note states: 'After you create instances, you cannot change the regions of the instances. Instances in different regions cannot communicate with each other over the internal network. Select a region that is close to your users to reduce network latency and accelerate data transmission.'

Network and Zone: Default VPC, Hong Kong Zone D | Default vSwitch. Links for 'Create VPC' and 'Create vSwitch' are provided.

Instances & Images: Recently Used Instance Types, All Instance Types.

Filter Condition: Select a vCPU..., Select a mem..., Search by instance type name. I/O Optimized. View more specification parameters.

Architecture: x86, Arm-based, GPU/TPGA/ASIC, ECS Bare Metal Instance. All Categories. High-performance Computing. All Categories.

Instance Family	Instance Type	vCPUs	Memory	GPU/TPGA	GPU Mem	Reference Price
Memory-optimized Type r6i	ecs.r6i.2xlarge	8 vCPU	64 GiB	-	-	\$341.96 USD /Month
Memory-optimized Type r6i	ecs.r6i.3xlarge	12 vCPU	96 GiB	-	-	\$512.95 USD /Month

Configuration Summary: Save as Launch Template | ...

- Billing Method: Subscription
- Region: China (Hong Kong)
- Zone: Hong Kong Zone D
- Network Type: VPC
- VPC: Default VPC
- vSwitch: Default vSwitch
- Instance Type: Universal Type u1 / ecs.u1-c1m2.large (2 vCPU 4 GiB)
- Image: Ubuntu 22.04 64-bit (Security Hardened)
- System Disk: Enterprise SSD (ESSD) 80 GiB. Release with Instance. Show More
- Quantity: 1
- Duration: 3 Months
- Auto-renewal: ☐ Enable Auto-renewal
- Instance Price: 0.00 USD. View Details
- ☐ ECS Terms of Service | Product Terms of Service
- Add to Cart, Create Order

ECS configuration on Alibaba Cloud is introduced, outlining key details such as billing methods and instance types necessary for efficient setup.

Monitoring ECS Instance Status

The screenshot displays the Alibaba Cloud ECS console interface. The top navigation bar includes the Alibaba Cloud logo, a search bar, and various service links like Expenses, ICP, Enterprise, Support, Tickets, and a user profile. The left sidebar contains a navigation menu with categories like Elastic Compute Service, Overview, Events, Tags, Troubleshooting, Applications, My Services, Instances & Images, Network & Security, Storage & Snapshots, and Deployment & Elasticity. The main content area shows the details for an ECS instance named 'launch-advisor-20241102', which is in a 'Running' state. The instance details are organized into sections: Basic Information, Configuration Information, and Alerts for Critical Events. The Basic Information section includes fields for Instance ID, Instance Name, Instance Status, Zone, Billing Method, and Health Status. The Configuration Information section includes fields for Instance Type, CPU and Memory, Public IP Address, Primary Private IP Address, Image ID, Operating System, VPC, vSwitch, Network Type, and Internet Bandwidth. The Alerts for Critical Events section shows 'No alerts for critical events'.

Basic Information

Field	Value
Instance ID	i-j6czzre0mde78q3889o Connect Reset Password
Instance Name	launch-advisor-20241102 ✎
Instance Status	Running Stop Restart
Zone	Hongkong Zone D
Billing Method	Subscription Switch to Pay-As-You-Go
Health Status	Normal

Configuration Information

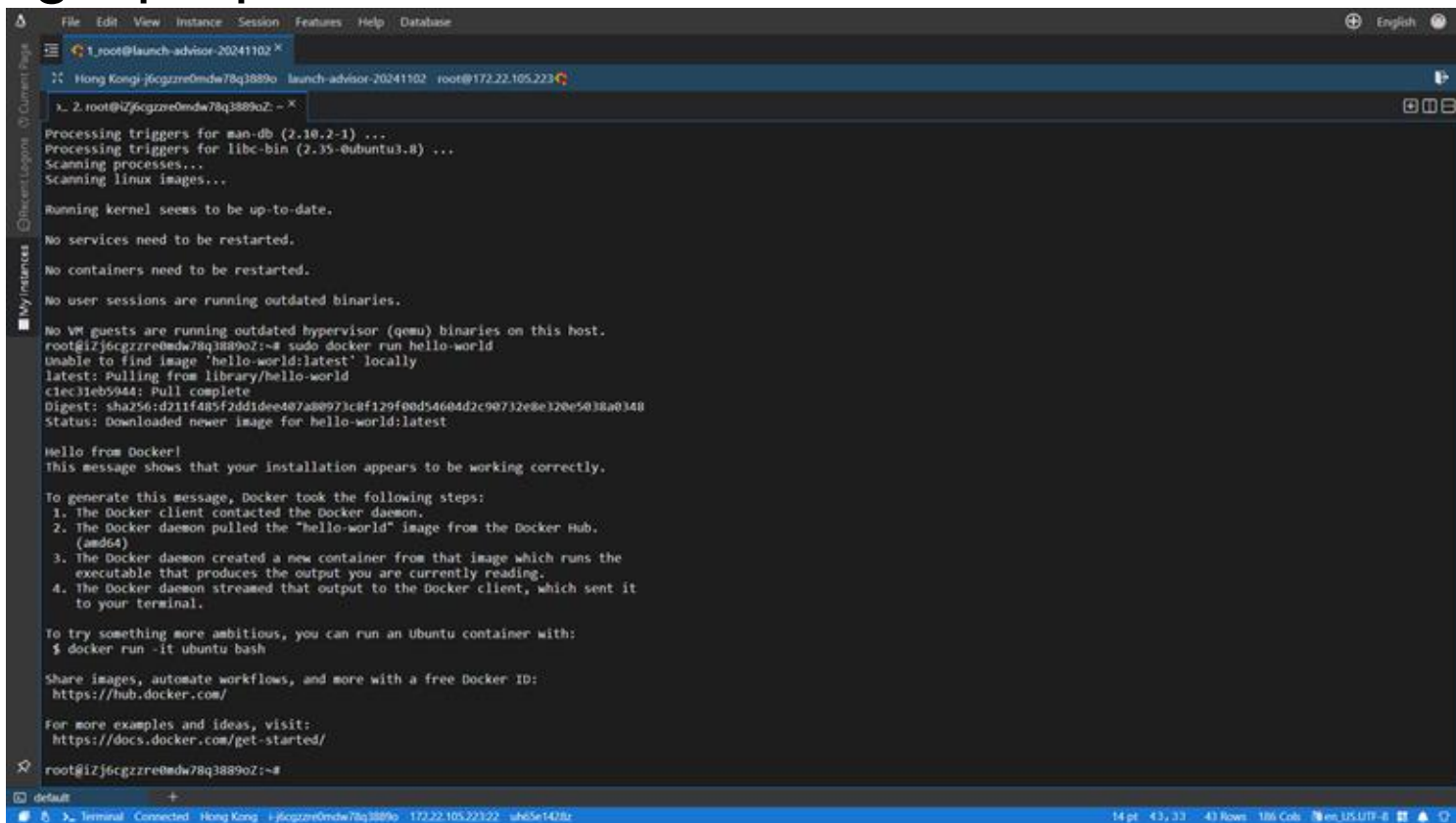
Field	Value
Instance Type	ecs.u1-c1m2.large ✎ Change Purchase Same Specifications
CPU and Memory	2 Cores (vCPU) 4 GiB
Public IP Address	47.243.80.49 ✎ Convert to EIP
Primary Private IP Address	172.22.105.223 ✎
Image ID	ubuntu_22_04_x64_20G_alibase_20240926.vhd ✎ Create Custom Image
Operating System	Ubuntu 22.04 64 bit Change
VPC	vpc-j6qjhwz37okaf3botrf ✎ Change
vSwitch	vsw-j6ceheg5csum2pys0q35 ✎ Change
Network Type	VPC
Internet Bandwidth	1 Mbps Change Subscription Instance Bandwidth

Alerts for Critical Events [View Details](#)

No alerts for critical events.

An overview of an ECS instance named "launch-advisor-20241102" is provided, detailing its current status and essential information for effective resource management.

Setting Up Apache Airflow with Docker



```
File Edit View Instance Session Features Help Database
1_root@launch-advisor-20241102 x
Hong Kong: j6cgzzre0mdw78q3889o launch-advisor-20241102 root@172.22.105.223
x. 2_root@iZj6cgzzre0mdw78q3889oZ: ~ x

Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.
No services need to be restarted.
No containers need to be restarted.
No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@iZj6cgzzre0mdw78q3889oZ:~# sudo docker run hello-world
unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
c1ec31eb5944: Pull complete
Digest: sha256:d211f485f2dd1dee407a0973c8f129f00d54604d2c90732e0e320e5038a0348
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

root@iZj6cgzzre0mdw78q3889oZ:~#
```

Terminal commands used to set up Apache Airflow with Docker are provided, showcasing how to verify a successful installation through a test case. The configuration of Apache Airflow is facilitated by utilizing Alibaba Workbench.

Setting Up Apache Airflow with Docker

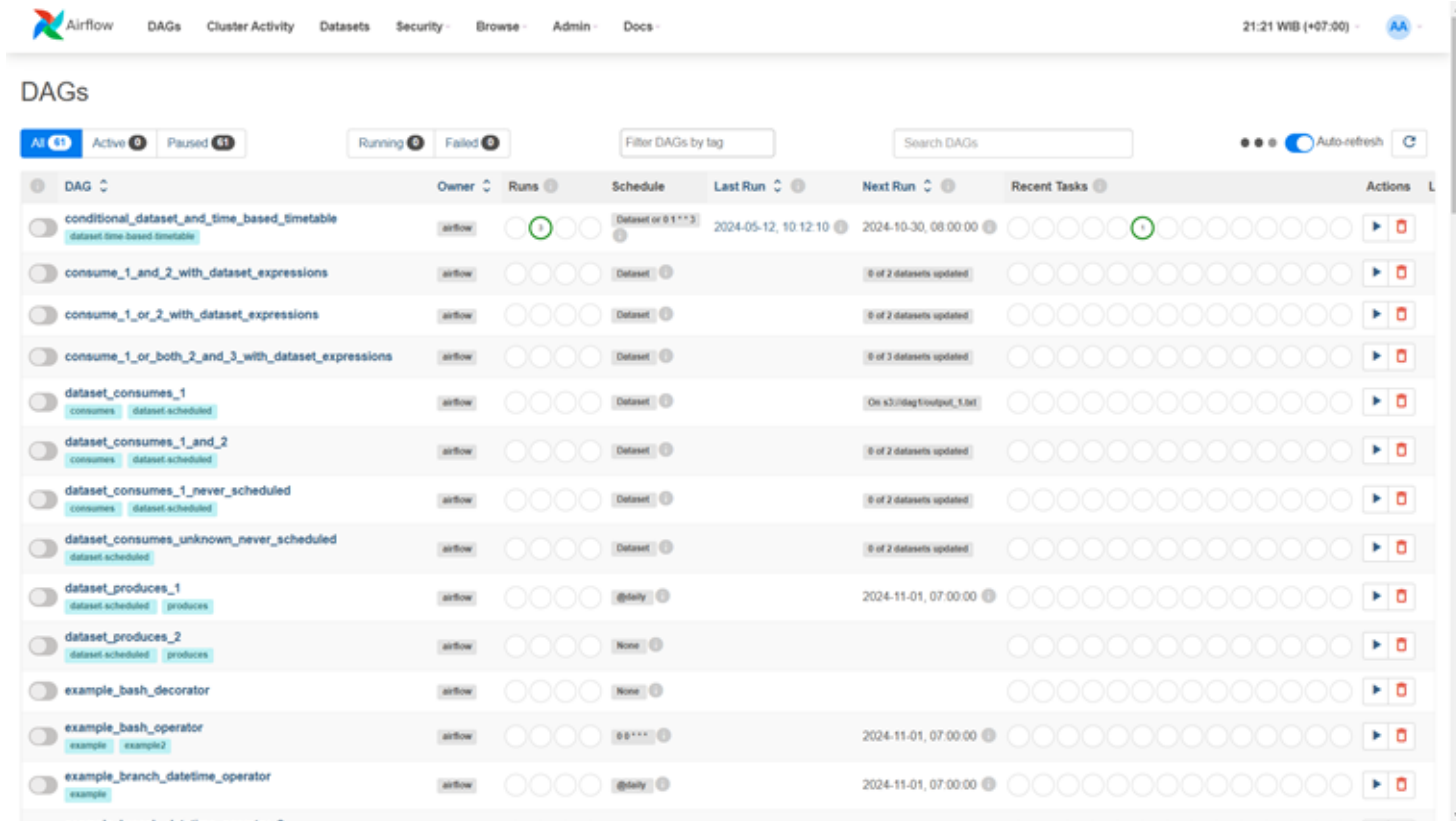
[illegible][illegible]

Logs and status messages related to initializing and running Apache Airflow services using Docker Compose.

For further installation details, refer to the official:

- Docker documentation for Ubuntu: <https://docs.docker.com/engine/install/ubuntu/>
- Apache Airflow Docker Compose guide: <https://airflow.apache.org/docs/apache-airflow/stable/howto/docker-compose/index.html>

Navigating Apache Airflow UI

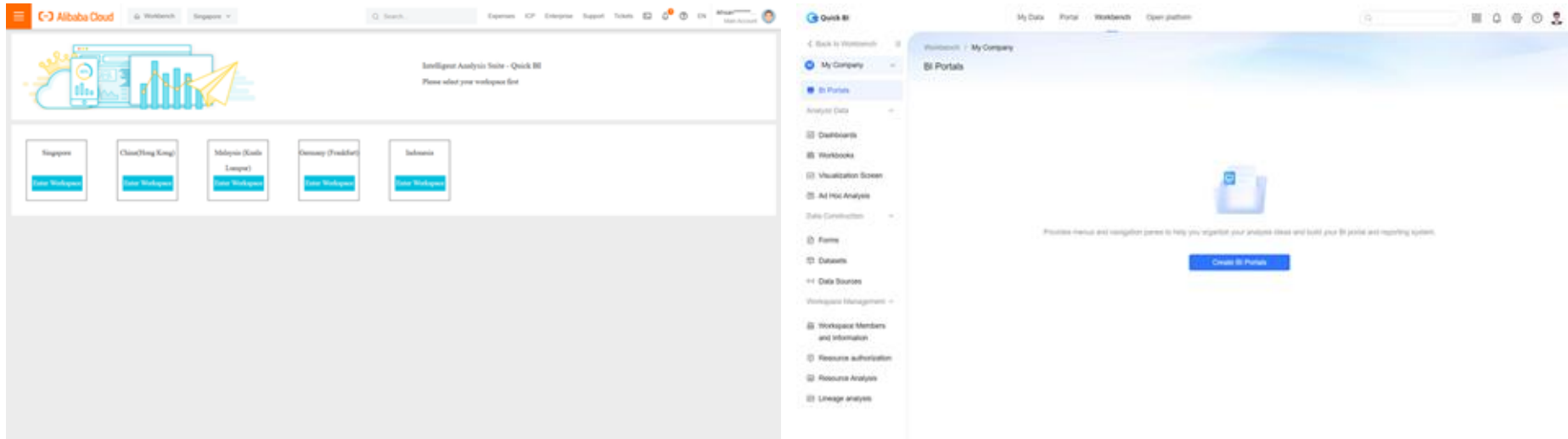


The screenshot displays the Apache Airflow web interface. At the top, there is a navigation bar with links for DAGs, Cluster Activity, Datasets, Security, Browse, Admin, and Docs. The current time is 21:21 WIB (+07:00). The main section is titled 'DAGs' and includes filters for 'All' (69), 'Active' (9), and 'Paused' (3). There are also buttons for 'Running' (3) and 'Failed' (3), a 'Filter DAGs by tag' input, and a 'Search DAGs' input. An 'Auto-refresh' toggle is set to 'On'. The table below lists various DAGs with columns for DAG name, Owner, Runs, Schedule, Last Run, Next Run, Recent Tasks, and Actions.

DAG	Owner	Runs	Schedule	Last Run	Next Run	Recent Tasks	Actions
conditional_dataset_and_time_based_timetable dataset time based timetable	airflow	1	Dataset or 0 1 * * *	2024-05-12, 10:12:10	2024-10-30, 08:00:00	1	▶ ⛔
consume_1_and_2_with_dataset_expressions	airflow		Dataset		0 of 2 datasets updated		▶ ⛔
consume_1_or_2_with_dataset_expressions	airflow		Dataset		0 of 2 datasets updated		▶ ⛔
consume_1_or_both_2_and_3_with_dataset_expressions	airflow		Dataset		0 of 3 datasets updated		▶ ⛔
dataset_consumes_1 consumes dataset scheduled	airflow		Dataset		On x3:0day1:00put_1.txt		▶ ⛔
dataset_consumes_1_and_2 consumes dataset scheduled	airflow		Dataset		0 of 2 datasets updated		▶ ⛔
dataset_consumes_1_never_scheduled consumes dataset scheduled	airflow		Dataset		0 of 2 datasets updated		▶ ⛔
dataset_consumes_unknown_never_scheduled dataset scheduled	airflow		Dataset		0 of 2 datasets updated		▶ ⛔
dataset_produces_1 dataset scheduled produces	airflow		daily		2024-11-01, 07:00:00		▶ ⛔
dataset_produces_2 dataset scheduled produces	airflow		None				▶ ⛔
example_branch_decorator	airflow		None				▶ ⛔
example_branch_datetime_operator example example2	airflow		0 0 * * *		2024-11-01, 07:00:00		▶ ⛔
example_branch_datetime_operator_2 example	airflow		daily		2024-11-01, 07:00:00		▶ ⛔

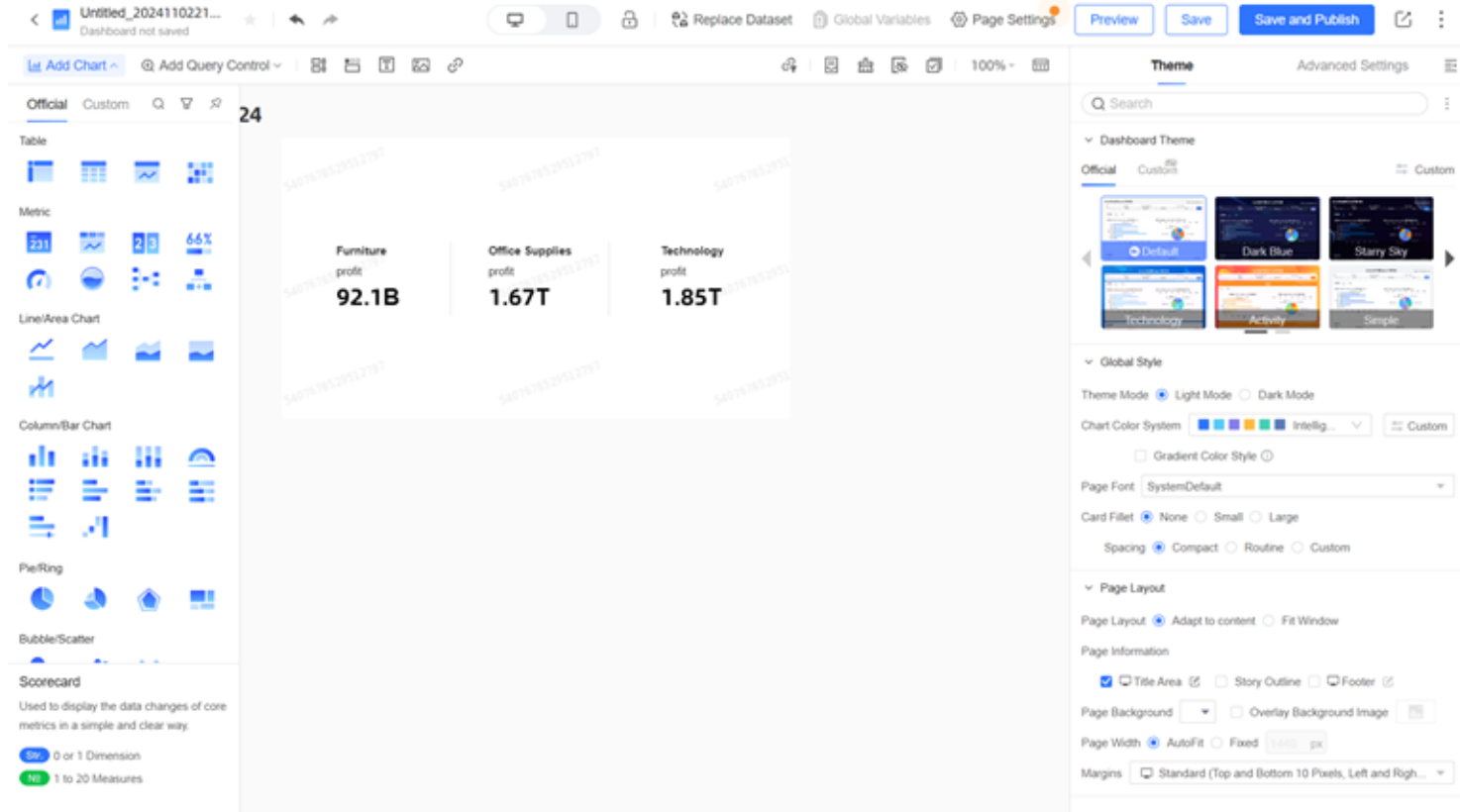
The Apache Airflow UI is displayed, where users can manage Directed Acyclic Graphs (DAGs), highlighting features that facilitate workflow management.

Quick BI Workspace Selection



The process of selecting a workspace for Quick BI on Alibaba Cloud is introduced, showcasing various geographical locations available for deployment.

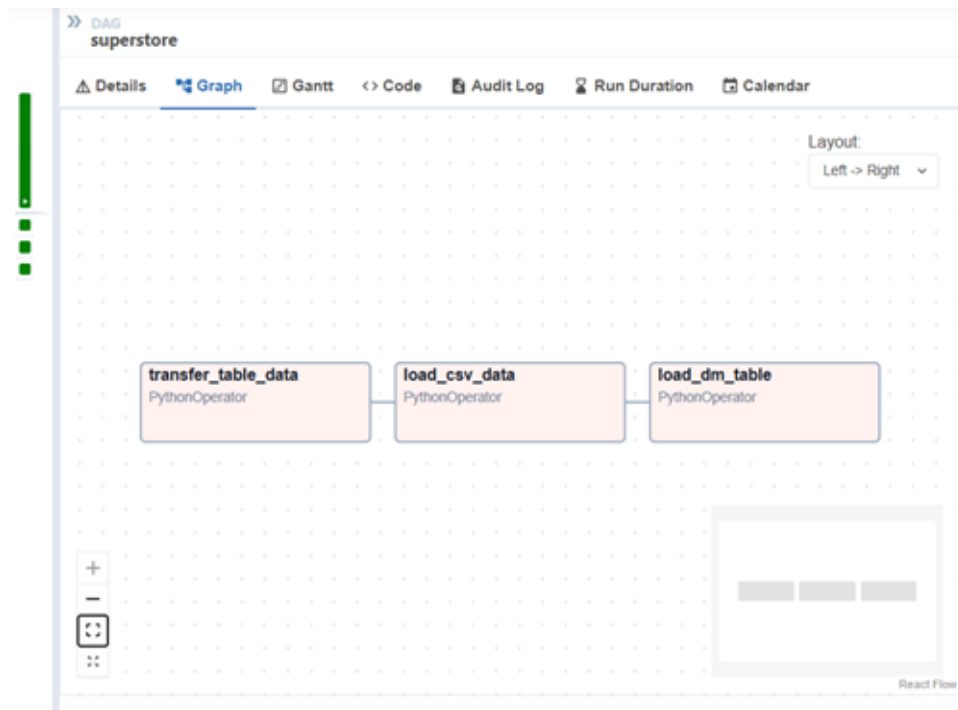
Customizing Quick BI Dashboards



The Quick BI interface used for dashboard creation is explored, highlighting various chart types and customization options to enhance data visualization.

Business Intelligence Implementation

Airflow DAG - Data Pipeline



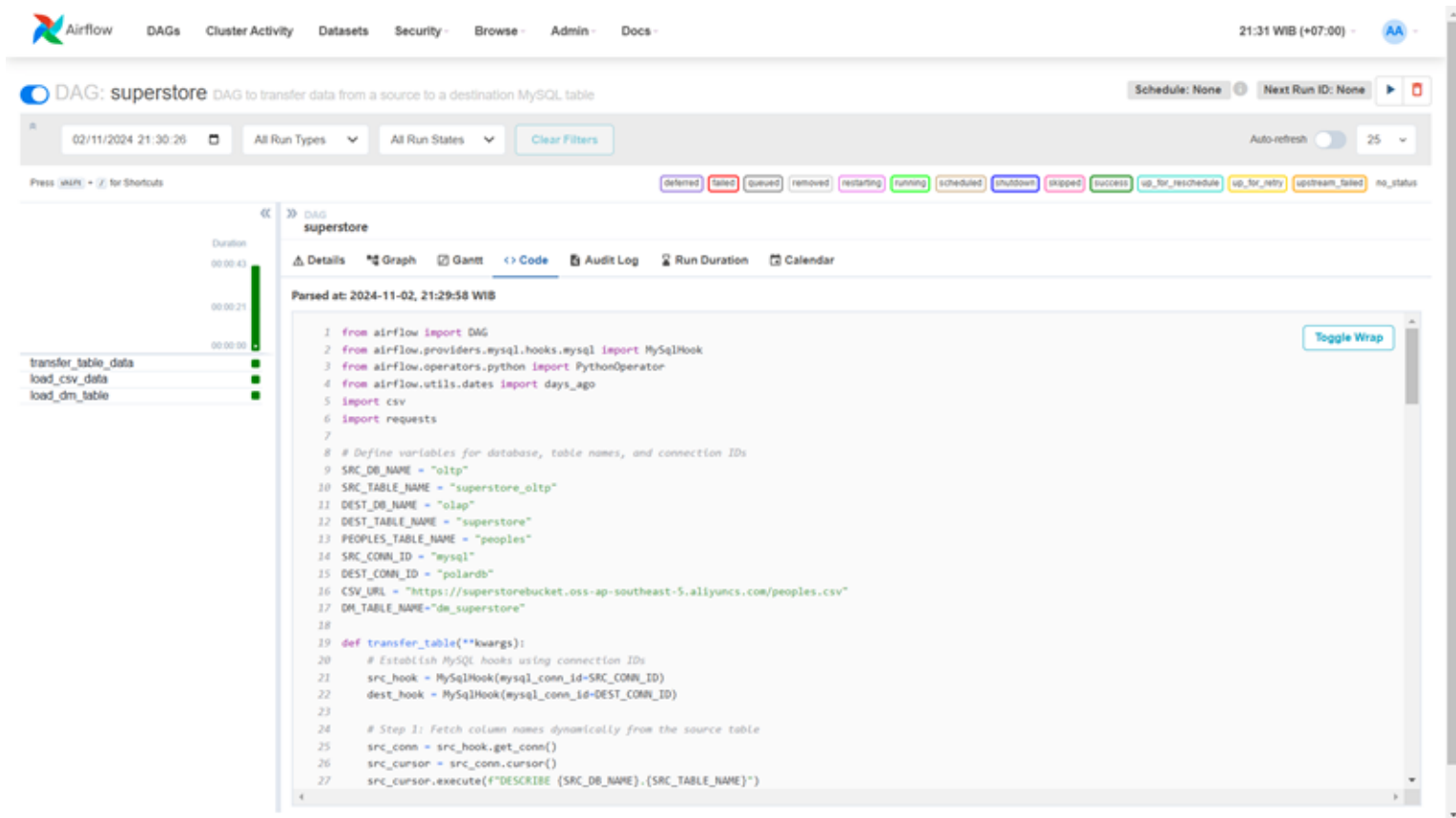
Data Transfer: The DAG transfers data from the MySQL source table (superstore_oltp) to the OLAP destination table (superstore), dynamically retrieving column names for efficient insertion.

CSV Ingestion: It fetches peoples.csv, clears the target peoples table in OLAP, then parses and inserts the new data.

Data Mart Update: The DAG then updates dm_superstore, truncating old data and inserting aggregated data from superstore and peoples to enhance BI analytics.

For more on DAGs, see the Airflow documentation here.
<https://github.com/ikhsannur1996/bi-alibaba-cloud/blob/main/superstore.py>

Airflow DAG - Data Pipeline



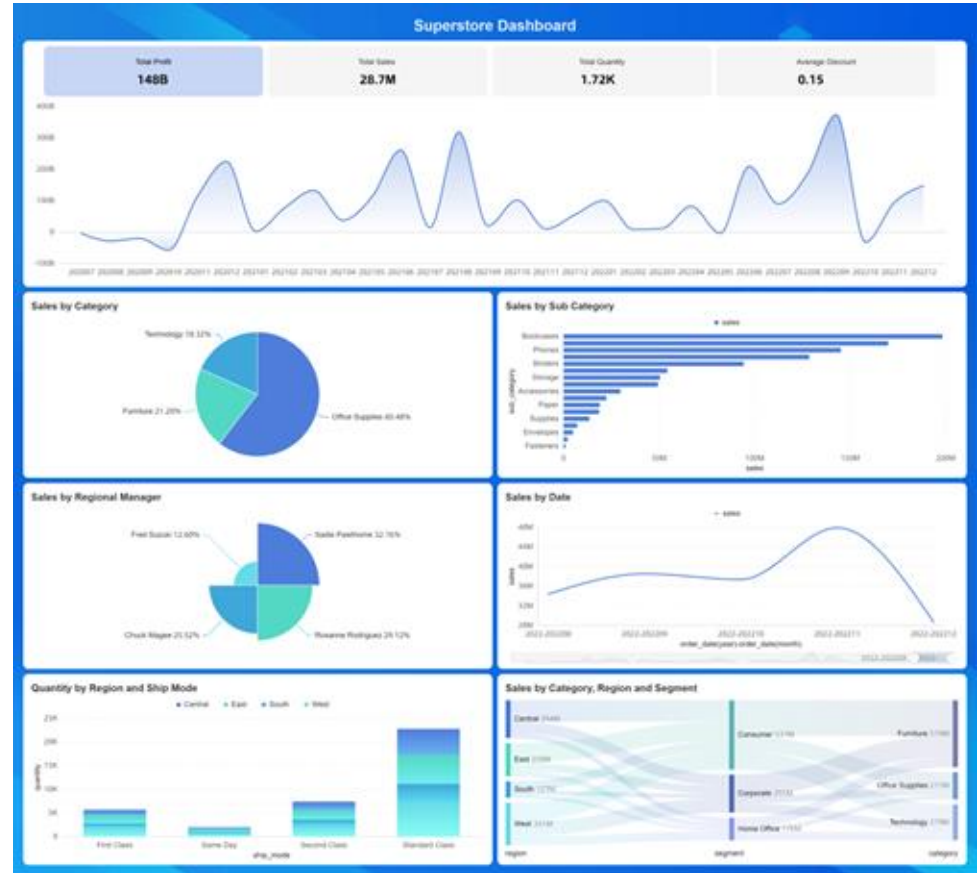
The DAG flow has been completed successfully, with each task marked in green.

Quick BI - Dashboard

The Quick BI dashboard provides a clear snapshot of superstore performance with:

- **Key Metrics:** Quick view of total profit, sales, quantity, and average discount.
- **Sales Trends:** Line graphs highlight peak sales periods.
- **Category Breakdown:** Pie and bar charts show sales by category and sub-category.
- **Regional Analysis:** Charts on sales by manager, region, and ship mode reveal regional performance and shipping efficiency.

For the full dashboard, see [here](#).



Conclusion and Recommendation

Conclusion

- **Scalable Infrastructure:** The combination of Quick BI, ECS, and ApsaraDB RDS offers a scalable platform for data analytics.
- **Efficient Deployment:** The architecture allows for quick setup and deployment of BI solutions.
- **Powerful Visualization:** Quick BI enables users to create insightful visualizations easily.
- **Integrated Benefits:** Utilizing Alibaba Cloud services ensures high availability, security, and performance.

Recommendations

- **Enhance Security:** Implement strong security measures using IAM roles and database encryption.
- **Regular Backups:** Set up automated snapshots in ApsaraDB RDS and PolarDB for regular backups and use monitoring tools to track performance.
- **Optimize Costs:** Continuously review resource usage to identify cost-saving opportunities, including reserved instances.
- **Continuous Updates and Training:** Keep all services updated and provide ongoing training to team members on using Quick BI effectively.

Thank You